

Organisation of the Organism

Organelles Found in both animal and plant cells.

1- Cytoplasm: Jelly-like structure that is enclosed by the cell membrane, contains the cell organelles.

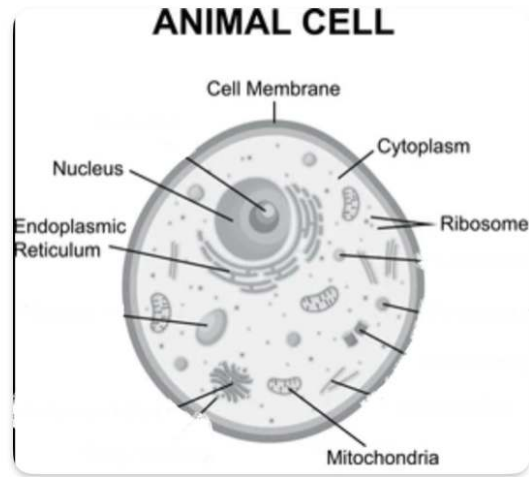
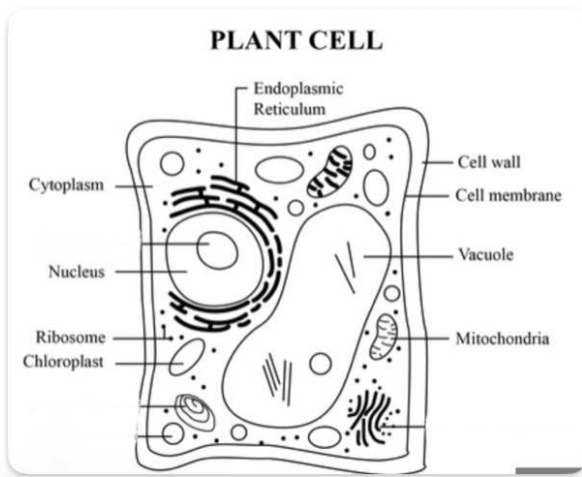
2- Cell membrane: Partially permeable layer that forms a boundary around the cytoplasm. Prevents cell organelles from escaping and controls what goes in and out of the cell.

3- Nucleus: Circular structure containing DNA in form of chromosomes. Found in cytoplasm. Controls the cell division, development and activities.

4- Mitochondria: Circular/Slipper shaped organelle. Found in the cytoplasm. Responsible for aerobic respiration

5- Ribosomes: Small circular structures attached to the membrane or lie free. Found in the cytoplasm. Responsible for protein synthesis

6- Endoplasmic reticulum: Transports ribosome



Organelles found only in plants

- 1- Cell wall: A tough-non living layer made of cellulose. Surrounds the cell membrane. Prevents the plant cell from bursting, and allows water and salts to pass
- 2- Vacuole: A fluid-filled space surrounded by a membrane. Contains salts and sugars, keep plant cells rigid and firm
- 3- Chloroplast: An organelle containing chlorophyll which traps light energy for photosynthesis

In animals, excess glucose is stored as glycogen in the cytoplasm

In plants, excess glucose is stored as starch in the chloroplast

In bacteria: They have cytoplasm, cell membrane, ribosomes, cell wall made of chitin, circular DNA: A single circular chromosome that controls the cell's activities, development and division. They also have plasmids: Which are circular structures that contain genes that carry genetic information to help the process of survival and reproduction of the bacterium.

Specialized Animal Cells

1- Ciliated epithelial cells: Found in the lining of the nose, windpipe, bronchi and oviducts

Function: Sweeping the mucus away from the lungs and towards the throat.

* Mucus: A thick fluid that traps dust and bacteria

* Cilia: Hair like structures produced by the cytoplasm.

Adaptation: They have cilia on their surface

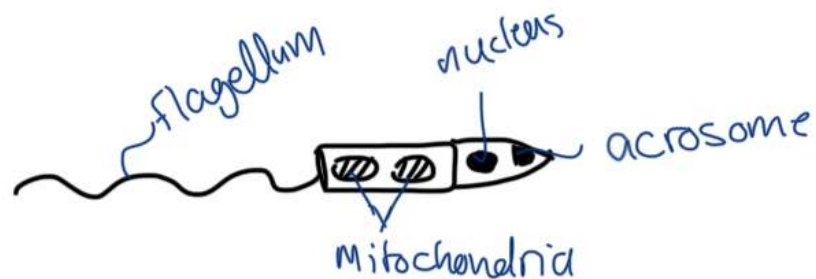
→ Smoking paralyses these cilia from working

2- Sperm cells: Called male gamete

Function: Fuse with egg cell to form a zygote
in the process of fertilization.

Adaptations:

- 1- Have a tail so that it can swim to reach the ovum
- 2- Has many mitochondria to produce a lot of energy to swim

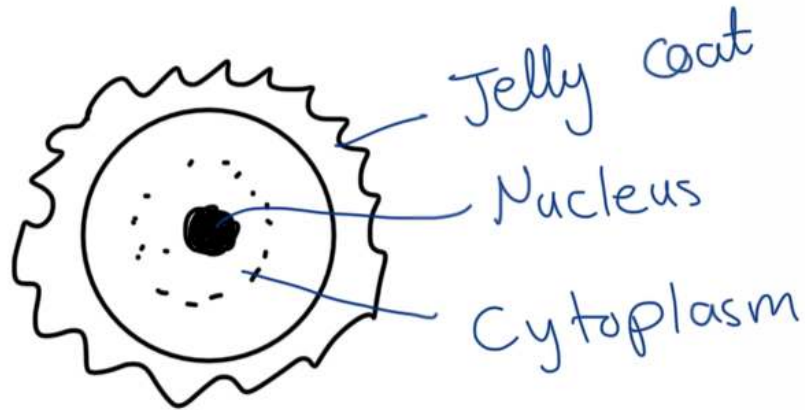


3- Egg cell: ovum: Called female gamet

Function: Fuse with the sperm cell to form a zygote during Fertilization

Adaptation:

- 1- Dense cytoplasm filled with nutrients
- 2- Has a jelly coat that hardens after fertilization to prevent Polyspermy



4-Red blood cells

Function: Transport oxygen around the body

Adaptations:

- 1- Have no nucleus to have more space to carry more haemoglobin
- 2- Have haemoglobin that is a red pigment that carries oxygen
- 3- They have a biconcave shape which can squeeze into blood capillaries and increase the surface area so that red blood cells carry more oxygen

5- White blood cells:

Function: To defend the body against disease

Adaptations:

1- Have nucleus

2- Can change shape to squeeze in blood vessels and engulf bacteria



Phagocyte:

Engulf bacteria by a process called phagocytosis.

Lymphocyte: Release anti-bodies

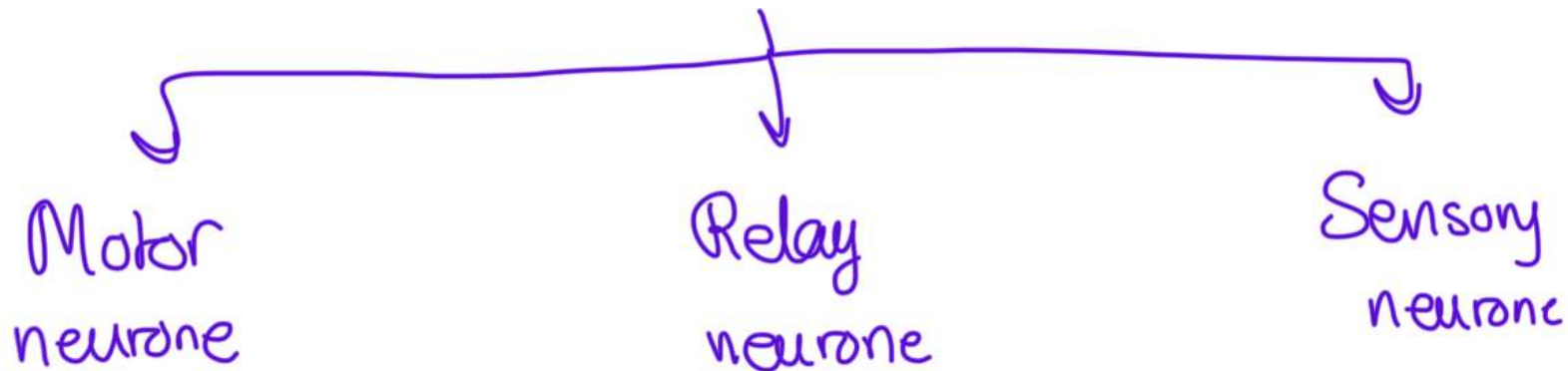
6- Nerve cells :- Which is present in the nervous system of animals

Function: Carry electrical impulses all around the body

Adaptations:

1- Have axons which are long extensions of cytoplasm that deliver impulses to other neurones.

2- Have myelin sheath around the axons to protect them and insulate them for faster transmissions of impulse



Specialized Plant cells

1- Palisade mesophyll cells: - Present in leaves

Function: Carry out most of the photosynthesis

Adaptations:

- * Have a lot of chloroplasts to capture a lot of light
- * Cells are tightly packed to capture as much light as possible.

2- Root hair cells : - Present in roots

Function: Absorb water and minerals from the soil.

Adaptations:

- * Have large surface area to absorb a lot of water
- * Have many mitochondria to release a lot of energy for absorption of minerals by active transport.

3- Xylem vessels : - Form the xylem tissue

- A hollow tube, has the function of transporting water and minerals from roots to leaves, and to support the plant.

Adaptations:

- * Cells that make the xylem have no living contents
- * End of cell walls are removed so water can move freely as one complete column.
- * It is hollow from the inside to allow water to move freely
- * Has thick walls to withstand high ^{water} ~~total~~ pressure.
- * Has lignin in the cellwalls to make it rigid and water-proof